Effective Central Charges across Clean, Disordered, and Monitored Criticality: Benchmarks and an Exploratory Learning-Induced Metal–Insulator Transition

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Universal finite-size terms provide stringent tests of numerical calculations at two-dimensional critical points, but disorder and monitored dynamics complicate both their interpretation and their uncertainty. We establish a reproducible validation hierarchy using three benchmarks and then apply it to an exploratory learning-induced metal–insulator transition. For the clean Ising model we obtain $c = 0.498739$ from Monte Carlo data and $c = 0.499424$ from the exact transfer matrix. For the random-bond Ising model on the Nishimori line and a weakly self-dual disordered model we find effective central charges $c_{\text{eff}} = 0.456469$ and 0.444107, respectively. At the candidate monitored transition $\phi/\pi = 0.30$, entanglement and Casimir fits instead yield 3.060740 and 12.579933. Their strong disagreement, together with an unbracketed transition and unstable anisotropy calibration, makes the monitored central-charge inference inconclusive. The comparison shows how validated benchmarks, resampling at the correct independence level, and explicit claim gates prevent precise-looking fits from being mistaken for universal results.

I. INTRODUCTION

The coefficient of a universal finite-size correction is one of the most economical fingerprints of a two-dimensional critical point. Conformal invariance relates the free-energy density on a periodic cylinder to the central charge [1, 2], while interval entanglement carries the same coefficient in a quantum ground state [3]. These relations are especially useful computationally: a single dimensionless number tests the microscopic implementation, the approach to scaling, and the analysis pipeline at once.

That economy can also be deceptive. A subleading L^{-2} term is small, correlated data do not supply as many independent observations as their row count suggests, and quenched disorder changes the object whose logarithm is averaged. In monitored or decohering systems an additional space–time anisotropy converts fitted amplitudes into a central charge only after a velocity calibration. A fit may therefore have a small regression error while its physical interpretation remains uncontrolled.

Here we organize four existing calculations into a common validation framework. The clean square-lattice Ising model [4] tests the entire Monte Carlo and fitting chain against both $c = 1/2$ and an exact finite-width calculation. The Nishimori random-bond Ising model [5, 6] tests quenched averaging and disorder-level resampling. A weakly self-dual disordered model tests whether those controls survive a less strongly constrained critical manifold. Only after these benchmarks do we analyze the learning-induced metal–insulator-transition candidate in a monitored class-DIII network [7]. Figure 1 summarizes this hierarchy.

Our main methodological result is deliberately conservative. The three benchmarks reproduce their reference values within their bootstrap uncertainties. The monitored calculation does not pass the same gates: the putative DIII transition is not bracketed by the available scan, the anisotropy estimate is unstable, and two nominal central-charge estimators disagree. We report the amplitudes and their conditional conversions, but not a universal central charge for that model.

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II. MODELS AND UNIVERSAL OBSERVABLES

A. Clean and random-bond Ising models

For Ising spins $s_i = \pm 1$ on a square lattice, the reduced Hamiltonian is

$$-\beta H = \sum_{\langle ij \rangle} K_{ij} s_i s_j. \tag{1}$$

The clean benchmark uses $K_{ij} = K_c = \frac{1}{2}\ln(1+\sqrt{2})$, whereas the quenched benchmarks draw bond variables once per disorder realization and hold them fixed during thermal sampling. On the $\pm J$ Nishimori line, the bond distribution and inverse temperature obey the Nishimori condition [5]. The weakly self-dual ensemble uses the self-dual parameter specified by the frozen production data.

For an infinitely long cylinder of circumference L, conformal invariance gives

$$f(L) = f_\infty - \frac{\pi c_{\text{eff}}}{6L^2} + a_4 L^{-4} + \cdots, \tag{2}$$

where f is the dimensionless free-energy density. In a unitary clean model $c_{\text{eff}} = c$; for a quenched random system it denotes the effective central charge extracted from the disorder-averaged free energy. The averaging order is essential: the quenched observable is proportional to $[\ln Z]_{\text{dis}}$, not $\ln[Z]_{\text{dis}}$.

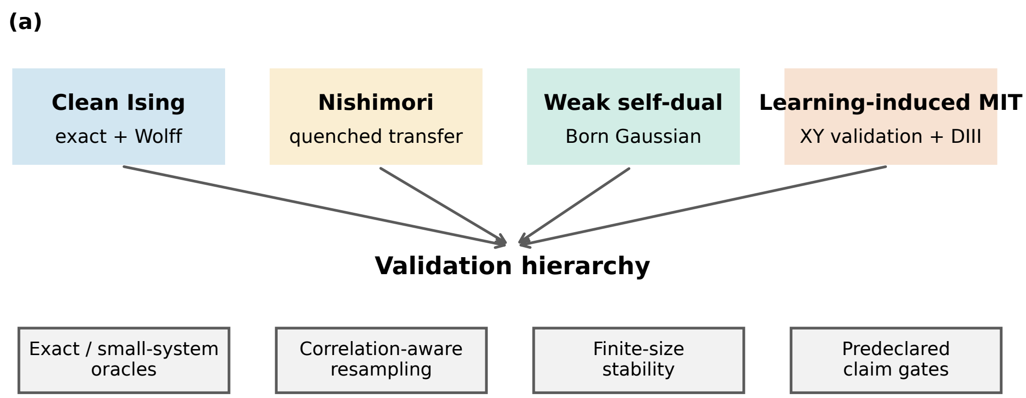

FIG. 1. Validation hierarchy used throughout this work. Exact and clean benchmarks validate normalization and scaling; disordered benchmarks add quenched averaging and common-disorder controls; the monitored calculation adds transition localization, anisotropy calibration, and agreement between independent universal estimators.

B. Monitored network model

The learning-induced problem is represented by a disordered free-fermion network in symmetry class DIII, with a measurement or decoherence angle ϕ controlling the trajectory ensemble. Its topological phases and possible intervening metallic regime are diagnosed by transfer-matrix localization observables and by the scale dependence of entanglement [7–9]. We analyze the frozen scan near $\phi/\pi = 0.30$ rather than retuning the candidate after inspecting its central-charge fits.

Two scaling relations are used. A periodic one-dimensional cut of length L has

$$S(\ell, L) = \frac{c_{\text{eff}}}{3} \ln\left[\frac{L}{\pi}\sin\left(\frac{\pi\ell}{L}\right)\right] + s_0 + \delta S, \qquad (3)$$

while the cylinder free energy follows Eq. (2) after converting the lattice aspect ratio with an anisotropy factor α. Agreement of these two estimates is a necessary internal consistency test; it is not imposed as a fitting constraint.

III. NUMERICAL METHODS AND VALIDATION HIERARCHY

A. Monte Carlo implementation

The Ising simulations are implemented in Rust. Every pseudorandom stream uses Xoshiro256++ [10], with seeds assigned before parallel execution so that scheduling cannot alter a realization. Thermal updates use the Wolff single-cluster algorithm [11]; measurements are separated by completed cluster updates, and warmup samples are discarded before accumulation. Energy-like observables are formed with a Rao–Blackwell estimator whenever the conditional bond expectation is available. This reduces thermal variance without changing the quenched sampling unit.

The simulation executable writes plain numerical records and metadata. Python is used only for validation, data reduction, nonlinear fits, bootstrap resampling, and publication graphics. This separation keeps the high-throughput stochastic kernel in a compiled language while making the statistical transformations directly inspectable.

B. Paired fitting and resampling

For each bootstrap replicate we resample independent units and repeat the full fit, rather than perturbing the final parameter covariance matrix. Clean thermal streams are the independent units. For random models a whole disorder realization—including every width measured from it—is resampled as a block. This paired bootstrap [12] preserves cross-width correlations. When two nearby parameter values are compared, the same bond realization and random labels are reused: this common-disorder construction reduces the variance of their difference without pretending that the pair is independent.

The reported standard error is the standard deviation of the bootstrap parameter distribution, and the interval is its central 95% percentile interval. Fit-window variation, truncation of small sizes, and correction terms are treated as systematic checks rather than folded into a single regression error. For the exact clean calculation no stochastic error is assigned.

TABLE I. The four calculations and the additional controls required as model complexity increases.

Model	Universal probe	Principal control
Clean Ising	Casimir term	Exact transfer matrix
Nishimori RBIM	Quenched Casimir term	Disorder bootstrap
Weak self-dual	Quenched Casimir term	Common disorder
Monitored DIII	Entanglement/Casimir	Claim gates

C. Predeclared claim gates

We distinguish a numerical fit from a physical claim. Benchmark values must be stable to the prescribed size cuts and statistically compatible with their reference targets. A monitored universal-central-charge claim additionally requires (i) a transition bracketed by independent phase diagnostics, (ii) a stable space–time anisotropy calibration, and (iii) mutually compatible entanglement and Casimir estimates. Failing a gate does not erase the computed result; it changes the allowed statement from a universal estimate to a diagnostic conditional on the assumed candidate point.

IV. BENCHMARK CENTRAL CHARGES

A. Clean Ising calibration

The clean calculation uses periodic $L \times M$ tori with $M = 8L$ and $L = 4, 6, \ldots, 20$. We obtain the free energy at K_c in two independent ways. The first evaluates the exact strip transfer matrix. The second integrates the measured energy from the known $K = 0$ normalization,

$$F(K_c) = -N \ln 2 + \int_0^{K_c} \langle H \rangle_K \, dK, \tag{4}$$

on a nested 129-point coupling grid. The shift between 65- and 129-point Simpson quadrature is 0.001955 in the final central charge, below the Monte Carlo standard error. The cluster streams pass first/second-half and between-replica tests with the predeclared $|z| < 4$ threshold.

Figure 2 shows that the stochastic and exact free energies share the same Casimir slope. The primary $L_{\min} = 6$ fit, including an L^{-4} correction, gives

$$\begin{aligned} c_{\mathrm{MC}} &= 0.498739 \pm 0.005225, \\ c_{\mathrm{exact}} &= 0.499424. \end{aligned} \tag{5}$$

The Monte Carlo 95% interval $[0.488719, 0.509064]$ contains 1/2. Diagnostic windows give $c_{\mathrm{MC}} = 0.499053$ for $L_{\min} = 4$ and 0.490505 for $L_{\min} = 8$; their increasing uncertainty and mutual compatibility show why the central window is used without selecting it a posteriori.

B. Nishimori random-bond Ising model

The $\pm J$ ensemble uses antiferromagnetic-bond probability $p = 0.1092212$ and $K_N = \frac{1}{2} \ln[(1-p)/p] = 1.0493604763$. For each of eight independent replicas, a matrix-free transfer product is propagated through 4096 burn-in rows and 2 097 152 measured rows at widths $L = 4, 6, 8, 10, 12, 14$. Normalizing after every row prevents overflow and turns the accumulated logarithmic norms into the quenched free-energy density.

The joint-width fit and hierarchical bootstrap in Fig. 3 give

$$\begin{aligned} c_{\mathrm{eff}} &= 0.456469 \pm 0.008181, \\ \mathrm{CI}_{95\%} &= [0.440064, 0.472304]. \end{aligned} \tag{6}$$

This interval contains the reference value 0.464 [6]. Removing $L = 4$ shifts the center from 0.456484 to 0.461475; a paired bootstrap of the difference contains zero. Independent microscopic audits also pass: the measured negative-bond fraction has $z = -0.524$, and a centered coupling derivative differs from the exact Nishimori energy identity by only 7.46×10^{-5}.

It is important to state the replica convention. Our value tests the *ordinary quenched* random-bond Ising free energy and therefore the reference $c_{\mathrm{eff}} \simeq 0.464$. The often-quoted value near 0.522 belongs to a different Born-weighted or higher-replica observable in monitored constructions. The two numbers are not competing estimates of the same quantity; substituting the higher-replica target into this benchmark would change the ensemble being measured.

C. Weakly self-dual Born ensemble

The third benchmark evolves a fermionic Gaussian covariance matrix at $\theta = \pi/4$ and $\beta = \beta' = 0.8813735870$. Sequential, state-conditioned Born sampling generates correlated outcomes according to

$$P(s|\Gamma) = \frac{1 + s \tanh(\beta) \langle i\gamma_a \gamma_b \rangle_\Gamma}{2}. \tag{7}$$

The leading Lyapunov or Shannon-free-energy rate is fitted over $L = 6, 8, \ldots, 32$ as $\gamma_1(L) = f_\infty L - \pi c_{\mathrm{eff}}/(6L) + a/L^3$. This Gaussian-state representation is consistent with standard efficient fermionic simulation methods [13].

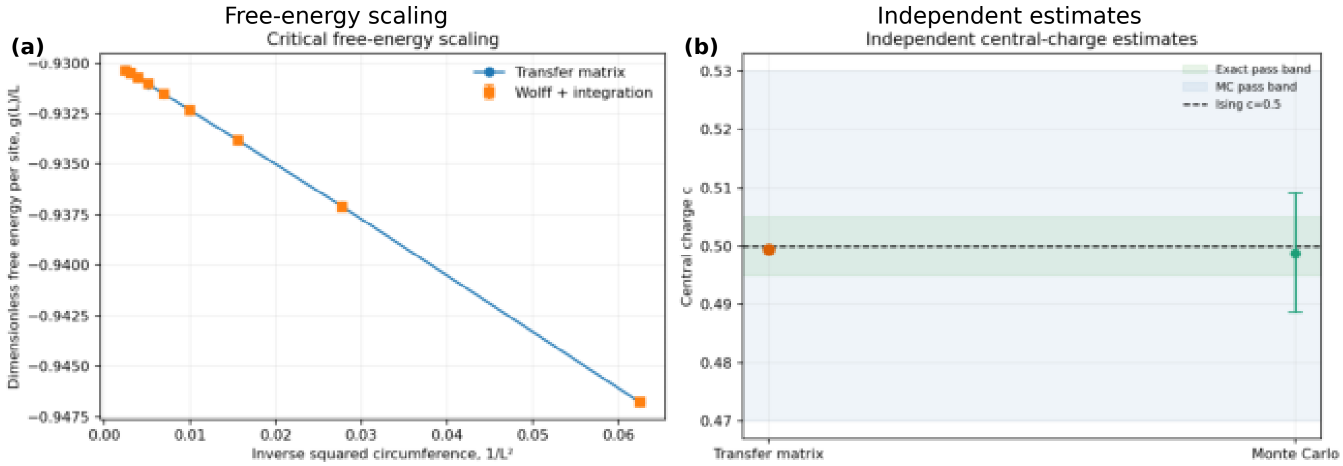

FIG. 2. Clean Ising calibration. (Left) Critical free-energy density versus L^{-2} from the exact transfer matrix and thermodynamic-integration Monte Carlo calculation. (Right) Independent estimates of c compared with the exact Ising value. Error bars are stream-level bootstrap uncertainties.

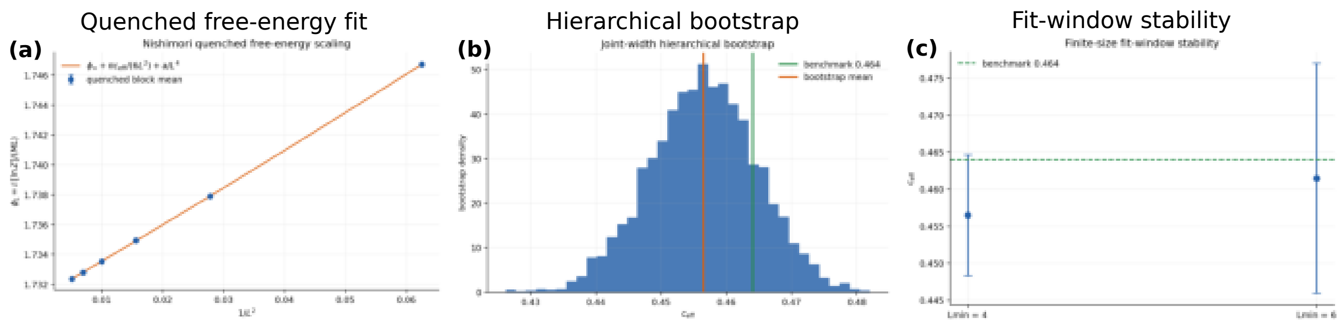

FIG. 3. Nishimori benchmark. (Left) Quenched free-energy density and the $L^{-2} + L^{-4}$ fit. (Center) Hierarchical bootstrap distribution of c_{eff}. (Right) Stability under the lower-width cut. A disorder realization is resampled together at all widths.

The result,

$$c_{\mathrm{eff}} = 0.444107 \pm 0.004240,$$
$$\mathrm{CI}_{95\%} = [0.435749, 0.452494], \qquad (8)$$

contains the reference 0.447. As shown in Fig. 4, the maximum studentized residual is 1.772, its correlation with $1/L$ is -0.056, the minimum effective sample size is 4517, and the electric–magnetic self-duality statistic is $z = -1.460$. Required fit variants span 0.00553, smaller than the declared 0.01 systematic tolerance.

The compact comparison in Fig. 5 makes the hierarchy visible: all three stochastic intervals cover their predeclared references, but the meaning of the coefficient changes from a unitary central charge to a quenched or Born-weighted effective central charge.

V. EXPLORATORY LEARNING-INDUCED METAL–INSULATOR TRANSITION

A. Phase localization and candidate selection

Measurement-induced entanglement transitions can exhibit conformal scaling, but localization, trajectory

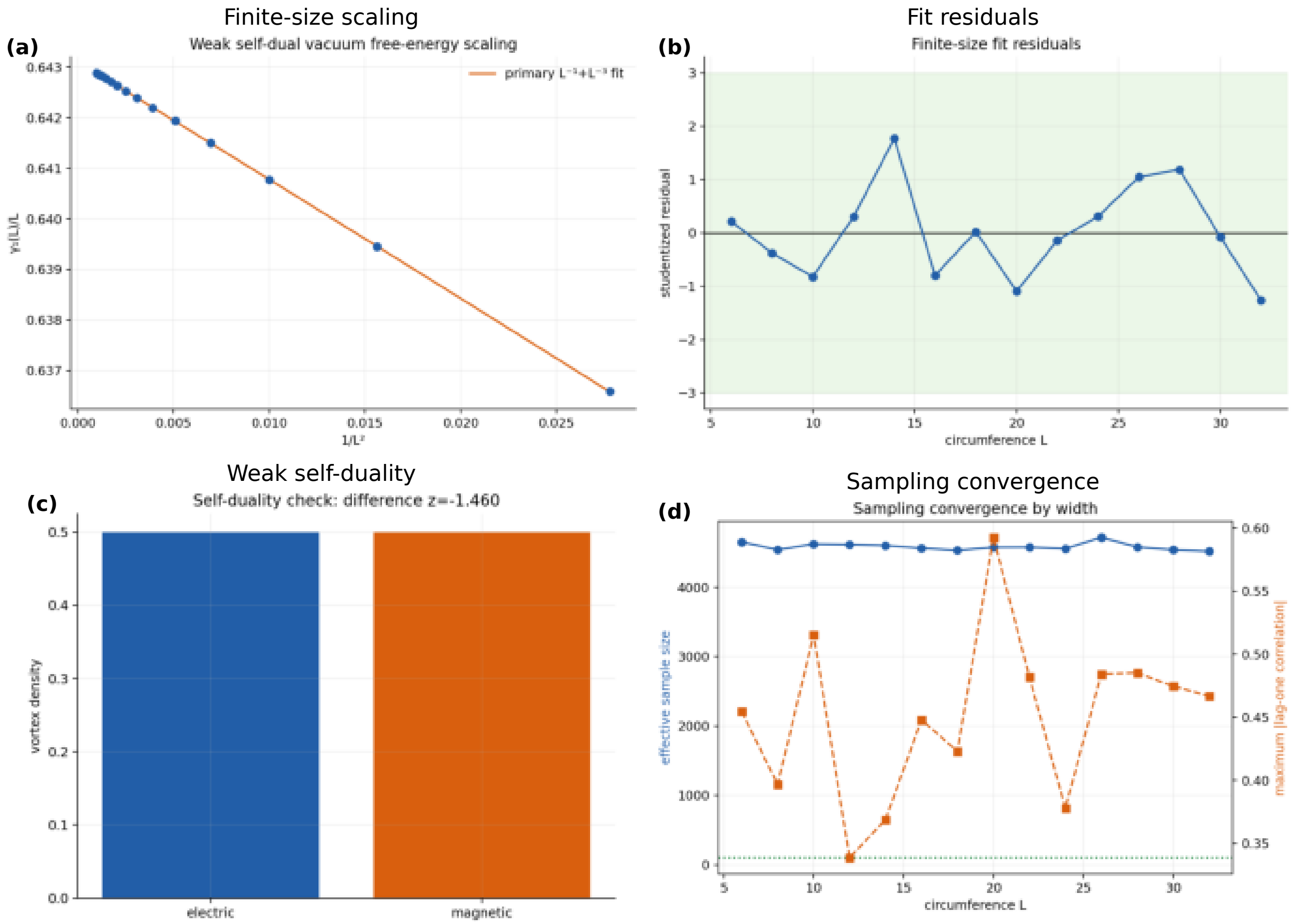

FIG. 4. Weak self-dual benchmark. The panels show the finite-size fit, studentized residuals, electric–magnetic self-duality audit, and autocorrelation/effective-sample-size convergence. The residual and self-duality diagnostics are evaluated independently of the target value.

TABLE II. Benchmark central charges. Parentheses denote one bootstrap standard error; intervals are central 95% bootstrap intervals. The exact clean result has no stochastic uncertainty.

Model	Widths	Estimate	95% interval	Reference	Outcome
Clean Ising, Monte Carlo	4–20	0.498739(0.005225)	$[0.488719, 0.509064]$	0.500	Pass
Clean Ising, exact	4–20	0.499424	—	0.500	Pass
Nishimori RBIM	4–14	0.456469(0.008181)	$[0.440064, 0.472304]$	0.464	Pass
Weak self-dual	6–32	0.444107(0.004240)	$[0.435749, 0.452494]$	0.447	Pass

weighting, and nonlinear replica limits all affect its interpretation [14, 15]. We therefore first validate the scanning observable in an XY slice. Its crossing lies in $\phi/\pi \in [0.24, 0.25]$, inside the reference window $[0.20, 0.28]$, as shown in Fig. 6. This confirms that the scan reacts correctly in the calibration geometry.

The generic DIII scan is different. Its diagnostic score increases monotonically from 0.023 at $\phi/\pi = 0.16$ to 1.387 at 0.32, but the available points do not exhibit both sides of a transition crossing. The transition is not bracketed. The boundary value $\phi/\pi = 0.30$ is retained as a preselected exploratory candidate, with the unobserved upper side extending to 0.32; it is not promoted to a critical point by the subsequent fits.

B. Entanglement estimate

At each width $L = 8, 12, \ldots, 32$, the interval entropy is fitted to the chord coordinate in Eq. (3). The width-

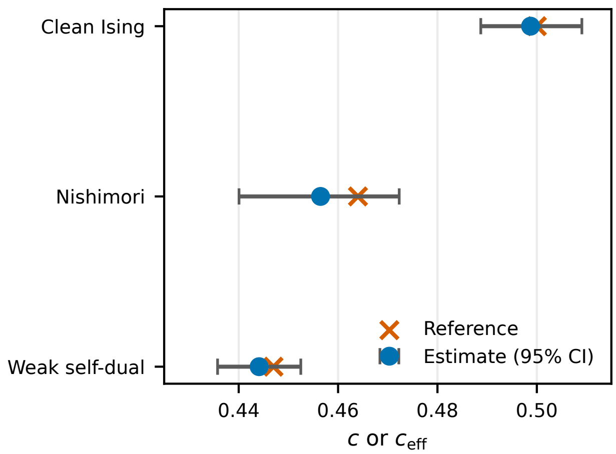

FIG. 5. Cross-benchmark comparison. Circles and horizontal bars show estimates and 95% intervals; crosses are reference values.

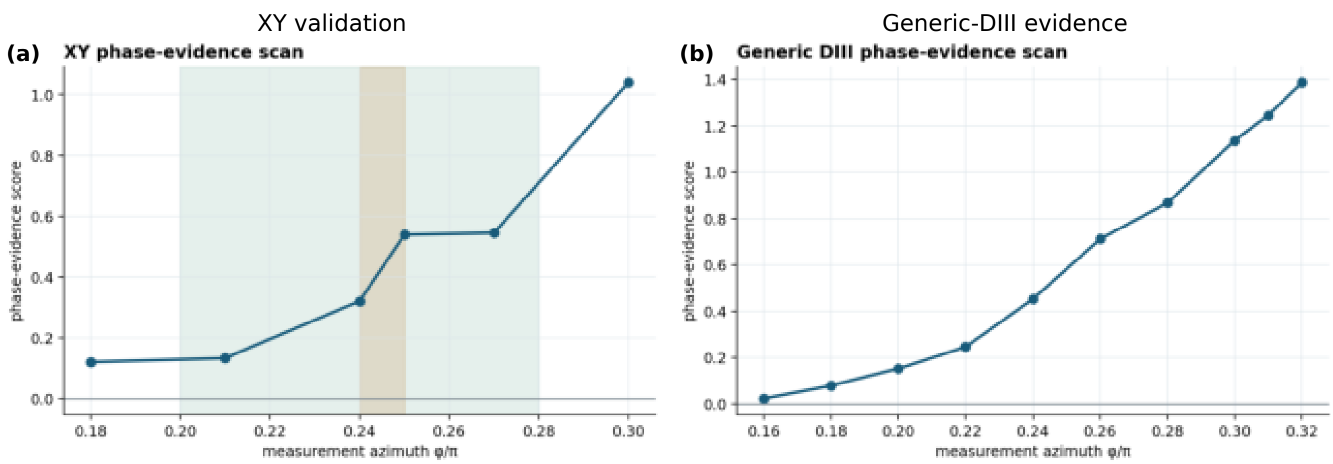

FIG. 6. Phase diagnostics. (Left) The XY validation scan brackets its transition between $\phi/\pi = 0.24$ and 0.25. (Right) The generic-DIII scan rises across the sampled interval but does not bracket a transition. The vertical marker at 0.30 is therefore an exploratory candidate rather than a localized critical point.

dependent coefficients are then extrapolated using a pre-specified model set; model weights are carried into the uncertainty rather than choosing the visually most favorable curve. Figure 7 shows both stages. The resulting conditional estimate is

$$\begin{aligned} c_{\mathrm{eff}}^{(S)} &= 3.060740 \pm 0.759748, \\ \mathrm{CI}_{95\%} &= [1.571634, 4.549845]. \end{aligned} \tag{9}$$

The fit has $\chi^2/\mathrm{dof} = 0.370$ and remains available after removing the smallest width, but the covariance condition number is 6.9×10^4. The wide interval correctly reflects noisy per-width coefficients rather than the smoothness of a single $L = 32$ chord fit.

C. Casimir estimate and failed consistency gates

The free-energy amplitude can be converted to a central charge only after calibrating the spatial-to-temporal anisotropy. The frozen calculation gives $\alpha = 0.027675$ with a nominal interval $[0.02584, 0.02956]$. However, its relative spread across admissible windows is 2.69 and the spatial/temporal decay mismatch is 0.958. Thus the anisotropy calibration is unstable even though a single-window regression error is small.

Using the nominal α gives the conditional Casimir

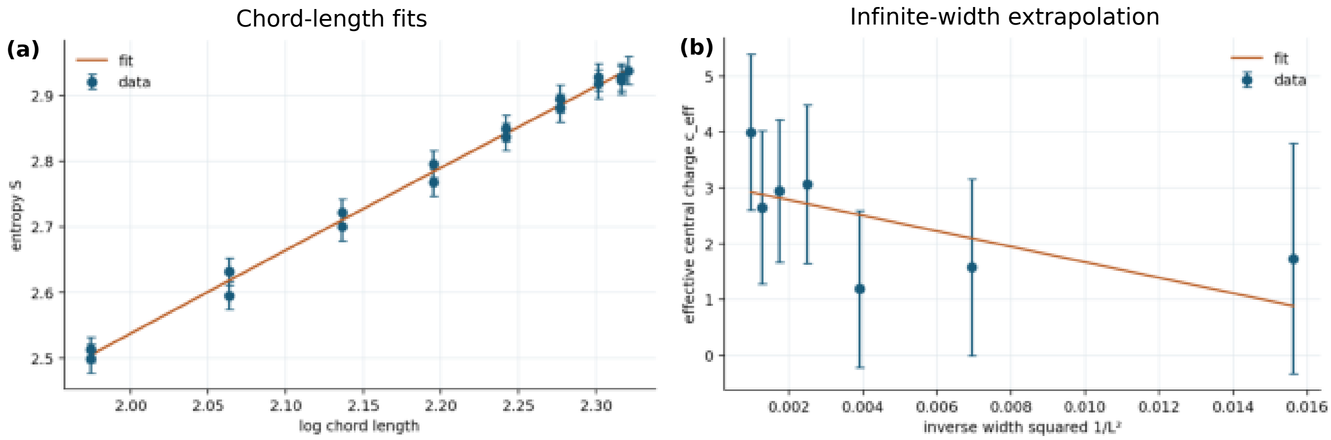

FIG. 7. Entanglement analysis at $\phi/\pi = 0.30$. (Left) Entropy versus conformal chord coordinate for the available widths. (Right) Extrapolation of the width-dependent effective coefficients. Model-selection uncertainty is included in the reported interval.

TABLE III. Conditional monitored-model results at $\phi/\pi = 0.30$. "No" means that the quantity is retained as a diagnostic but not published as a universal central charge.

Quantity	Value	Gate
XY bracket	$[0.24, 0.25]$	Pass
DIII bracket	None	Fail
$c_{\rm eff}^{(S)}$	3.060740(0.759748)	Available
α	0.027675	Unstable
$c_{\rm eff}^{(F)}$	12.579933(0.989806)	Available
Estimator agreement	No	Fail
Published central charge	false	Withheld

value

$$c_{\rm eff}^{(F)} = 12.579933 \pm 0.989806,$$
$$\mathrm{CI}_{95\%} = [10.727804, 14.610337]. \tag{10}$$

The difference $|c_{\rm eff}^{(F)} - c_{\rm eff}^{(S)}| = 9.519$ exceeds the combined 95% compatibility threshold 2.446: the estimators disagree. Figure 8 displays the apparently stable Casimir regression next to the unstable calibration and the final inconsistency. These diagnostics show why increasing Monte Carlo steps alone would not resolve the present problem. More steps reduce sampling variance, whereas the leading failures are an unbracketed parameter range, calibration drift, and estimator bias or mismatch.

Consequently, this pair of numbers in Table III does not constitute a universal-central-charge estimate. They are conditional diagnostics at an exploratory parameter value. This claim boundary is part of the result, not a qualification added after observing disagreement.

VI. CROSS-MODEL DISCUSSION

A. What the benchmark sequence validates

The common analysis separates three questions: whether a numerical observable is stable, whether its statistical error is correctly estimated, and whether the observable supports the intended universal interpretation. The clean model tests normalization, thermodynamic integration, the sign and factor of the Casimir coefficient, and finite-size corrections. Exact agreement rules out a broad class of coding errors that an internal goodness of fit cannot detect.

The Nishimori calculation then changes the statistical unit. Rows from one transfer product are correlated, and widths sharing a disorder realization are paired. Resampling rows or widths independently would underestimate the slope uncertainty. The identity and bond-frequency audits test the disorder generator separately from the fit. The weak self-dual model adds state-conditioned Born sampling: Gaussian invariant errors below 2.6×10^{-13} and the electric–magnetic check show that the measured coefficient is not merely a stable output of a drifting covariance matrix.

B. Error budget and parameter meaning

Table IV distinguishes error sources that are often collapsed into one bar. Statistical uncertainty is reduced by additional independent streams or disorder realizations. Autocorrelation is reduced by longer runs only until the effective sample size becomes adequate. Finite-size bias requires larger L or controlled correction terms.

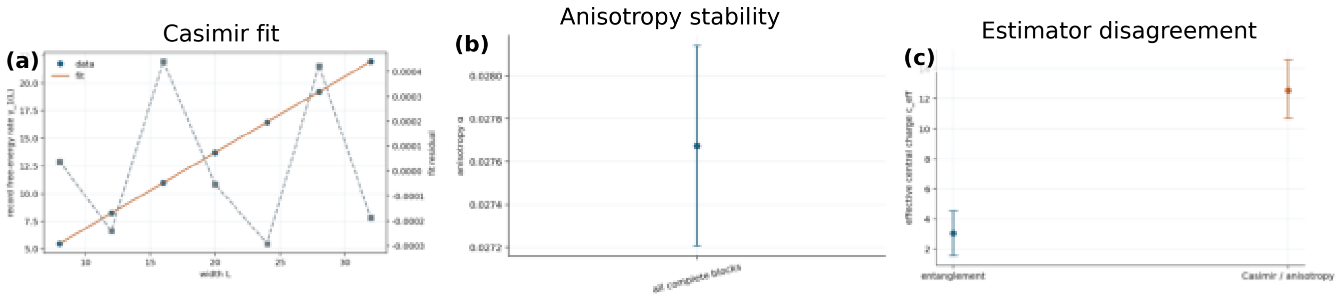

FIG. 8. Learning-induced-transition diagnostics. (Left) Casimir-amplitude fit at the exploratory candidate. (Center) anisotropy estimates under calibration variations. (Right) entanglement and nominally converted Casimir estimates, whose 95% intervals do not overlap.

Transition-location and anisotropy errors require new parameter coverage and calibration geometries; repeating the same point more accurately cannot remove them.

The large monitored values are therefore informative without being universal. They may reflect evaluation away from criticality, an incorrect anisotropic conversion, large finite-width corrections, or genuinely different replica observables. The present data do not discriminate among these mechanisms. In particular, the success of a smooth Casimir fit cannot substitute for agreement with entanglement or for a bracketed transition.

C. Next decisive calculation

The efficient next step is not uniform growth of the existing sample count. First, extend the DIII scan beyond 0.32 until the localization diagnostic shows a two-sided crossing, then refine that interval using common random numbers. Second, calibrate α from several aspect ratios at the bracketed point and require stability under width cuts. Third, recompute both universal estimators with larger widths and bootstrap their difference using shared disorder. Only if the difference interval contains zero should a combined central-charge estimate be quoted. This staged allocation directs computation at the dominant uncertainty rather than at the smallest error bar.

VII. CONCLUSION

We have placed four finite-size calculations in one PRB-style validation framework. Clean Ising thermodynamic integration yields $c = 0.498739(0.005225)$ and agrees with the independent exact value 0.499424. Realization-level resampling gives $c_{\rm eff} = 0.456469(0.008181)$ for the ordinary quenched Nishimori model, and state-conditioned Born sampling gives $c_{\rm eff} = 0.444107(0.004240)$ for the weakly self-dual benchmark. Each reference lies inside its 95% interval and independent implementation gates pass.

At the learning-induced metal–insulator-transition candidate, entanglement and nominal Casimir conversions produce 3.060740(0.759748) and 12.579933(0.989806), respectively. Because the DIII transition is unbracketed, the anisotropy is unstable, and the estimators are incompatible, false is correctly recorded as the publication status of a central charge. The defensible result is a reproducible failure of the universality gates together with a concrete refinement path. Across all four models, the central lesson is that more Monte Carlo steps improve accuracy only when sampling noise is the limiting error; they cannot repair an unlocalized critical point or an uncontrolled observable mapping.

Appendix A: Observable mappings

For a periodic cylinder with transfer direction $M \gg L$, write $-\ln Z = Mg(L)$ and $f(L) = g(L)/L$. Conformal

TABLE IV. Error mechanisms, controls, and effective remedies.

Source	Diagnostic	Present control	Remedy if failed
Thermal variance	replica/half-chain z	Wolff streams, blocking	more independent clusters
Quenched variance	bootstrap spread	realization-level paired bootstrap	more disorder realizations
Autocorrelation	ESS	block analysis	longer streams or better updates
Finite-size bias	$L_{\min}$ drift/residuals	L^{-4} or L^{-3} terms	larger widths
Quadrature bias	nested-grid shift	65/129-point comparison	denser coupling grid
Transition location	phase bracket	XY positive control	extend/refine DIII scan
Anisotropy	window spread	spatial/temporal comparison	dedicated aspect-ratio calibration
Estimator interpretation	S–F difference	independent estimators	resolve mapping before universality claim

invariance gives

$$g(L) = f_\infty L - \frac{\pi c}{6L} + O(L^{-3}), \tag{A1}$$

which is Eq. (2) after division by L. This derivation fixes both the sign and the factor of six. The clean transfer matrix supplies an independent microscopic mapping to the same coefficient.

For the $\pm J$ model, $g(L)$ is obtained from the long-product Lyapunov exponent. The physical quenched object is

$$f_q(L) = -\lim_{M\to\infty} \frac{[\ln Z_{L,M}]_{\rm dis}}{LM}. \tag{A2}$$

The sign convention of the stored logarithmic norm is converted once, before the finite-size fit. For the Born ensemble, $\gamma_1(L)$ is extensive in L, so the universal correction is L^{-1} rather than L^{-2} until the rate is divided by the width.

For monitored entanglement, the coefficient of the chord logarithm is first estimated separately at every width. The reported $c_{\rm eff}^{(S)}$ is the large-L extrapolation of those coefficients. The Casimir conversion contains the independently calibrated α; therefore its uncertainty and stability are logically prior to interpreting $c_{\rm eff}^{(F)}$.

Appendix B: Bootstrap construction

Let y_{rL} denote the estimate for realization or stream r at width L. A bootstrap replicate draws indices $r_1^*, \ldots, r_R^*$ with replacement and forms the complete vector

$$\bar{\boldsymbol{y}}^* = \frac{1}{R}\sum_{j=1}^{R} (y_{r_j^* L_1}, \ldots, y_{r_j^* L_n}). \tag{B1}$$

The nonlinear finite-size fit is repeated on this vector. All widths from one quenched realization consequently enter together, preserving the empirical cross-width covariance. The clean analysis applies the same construction to independent Monte Carlo streams. The Nishimori and weak-self-dual production analyses use 4000 replicates with fixed analysis seeds, but the simulation RNG streams remain independent of those seeds.

For common-disorder comparisons, parameter values a and b share the same bootstrap index in every replicate. The sampled quantity is $\Delta c^* = c_a^* - c_b^*$, not the difference of two independently resampled fits. This is the correct uncertainty for a paired fit-window or parameter comparison. Percentile intervals are read directly from the replicate distribution; no Gaussian shape is assumed.

Appendix C: Claim-gate audit

The frozen run contains 8 independent random streams and widths $8, 12, \ldots, 32$, with elapsed time 4583.689 s. Its summary artifact has SHA-256 digest

`cc08a6e6d6d414046c744b4d29d48f112d44526dfc2145b867aae01f07d53c33`.

The claim gate evaluates three Boolean conditions. First, the DIII phase diagnostic must supply a lower and an upper boundary; it supplies neither a complete bracket nor a crossing in the available interval. Second, the anisotropy must be stable across prescribed windows; its relative spread is 2.69. Third, the independent 95% intervals for the entanglement and Casimir conversions must be compatible; their center difference 9.519 exceeds the combined threshold 2.446.

All three failures are recorded in the source summary as `diii_transition_not_bracketed`, `anisotropy_unstable`, and `estimator_disagreement`.

No post-processing rule overrides them. The machine-readable publication flag is false. This audit ensures that a reader can reproduce not only the numerical values but also the logical reason they are not presented as a central charge.

1. Reproduction workflow

The analysis uses immutable result directories containing configuration, manifest, raw records, processed summaries, and plots. Rust executables generate stochastic records with Xoshiro256++ streams; Python scripts load hash-gated artifacts, perform resampling and fitting, and build the figures. The manuscript values are generated from those adapters rather than copied by hand. The complete workflow is invoked by the package Makefile, which runs tests, regenerates figures and TeX macros, compiles the RevTeX source, and verifies the resulting PDF.

DATA AVAILABILITY

All numerical results used in this article are retained in the accompanying repository as hash-identified result artifacts. Each artifact contains its configuration, manifest, raw or block-level records, processed summary, and analysis figures. No external proprietary data are required.

AUTHOR CONTRIBUTIONS

Xu Tian designed and implemented the numerical calculations, performed the analysis, and drafted the manuscript. Huidan Tan contributed to the physical interpretation, validation strategy, and revision of the manuscript. Both authors reviewed and approved the final paper.

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